

Part 3 - Sales Challan Workflow

The heart of the assignment: challan creation, draft/confirm lifecycle, stock validation and historical snapshots.

How to read this pack

Blue boxes are explicit case-study requirements recovered from the source explanation. Green boxes are implementation recommendations or inferences. This distinction helps you explain what was mandatory and what you chose as a developer.

Language style: simple Hinglish, interview-focused, and implementation-oriented. The goal is to help you build the project and confidently explain why each part exists.

1. Why Sales Challan Is The Core Workflow

Sales Challan wholesale/distribution business ka practical sales document hai. Customer selected hota hai, multiple products add hote hain, quantities define hoti hain, challan number generate hota hai, and confirmation ke baad stock reduce hota hai.

Case-study requirement

Challan must allow customer selection, multiple products, automatic challan number, draft/confirmed/cancelled statuses, stock reduction on confirmation, no negative stock, proper error on insufficient stock and product snapshot data in challan items.

2. Challan Creation Steps

- 1 Sales user selects customer, for example ABC Traders.
- 2 Sales user adds one or more products, for example 5 Monitors and 10 Keyboards.
- 3 System calculates total quantity and amount from line items.
- 4 System creates or assigns a unique challan number such as CH-2026-0001.
- 5 User saves as Draft first. Stock does not reduce yet.
- 6 User clicks Confirm. Backend validates stock and then reduces stock inside a transaction.

Customer: ABC Traders

Items:

Monitor quantity 5

Keyboard quantity 10

Status: DRAFT initially

Stock change: none until CONFIRMED

3. Automatic Challan Number

Challan number manual nahi hona chahiye, warna duplicate or inconsistent numbers aa sakte hain. System generated number better hai: CH-2026-0001, CH-2026-0002, etc.

Recommended implementation

Use a database unique constraint on challan_number. Generate with year + sequence. For a simple 48-hour build, a timestamp/sequence hybrid is acceptable if uniqueness is guaranteed.

4. Draft vs Confirmed

Ye assignment ka sabse important business concept hai. Draft ka matlab challan prepare ho raha hai, sale final nahi hui. Confirmed ka matlab transaction final ho gayi. Stock sirf confirmed hone par reduce hona chahiye.

Status	Meaning	Stock impact
DRAFT	Work in progress; editable	No stock reduction
CONFIRMED	Finalized sales challan	Stock reduced and OUT movement created
CANCELLED	Challan no longer active	Usually no stock change if cancelled from draft

5. No Negative Stock

Agar product stock 5 hai aur customer 10 maang raha hai, backend ko transaction reject karni chahiye. Stock -5 nahi hona chahiye. User ko clear error milna chahiye: available 5, requested 10.

```
Available stock = 5
Requested quantity = 10

Result:
409 Conflict
{ message: 'Insufficient stock', available: 5, requested: 10 }
```

Interview point

Stock validation must happen in backend, ideally inside a database transaction. Frontend validation improves UX, but it cannot be trusted as the final business rule.

6. Confirm Challan Transaction

```
BEGIN TRANSACTION

Load challan with items
Check challan status is DRAFT
For each item:
  lock/read product stock
  if stock < requested quantity:
    ROLLBACK and return 409

For each item:
  reduce product current_stock
  create stock_movement OUT with reason Sale Challan

Update challan status to CONFIRMED
COMMIT
```

Transaction isliye zaroori hai kyunki partial update dangerous hai. Agar Monitor stock reduce ho gaya but Keyboard update fail ho gaya, data inconsistent ho jayega. Transaction all-or-nothing behavior deta hai.

7. Product Snapshot Concept

Case-study requirement

Challan should store product snapshot data, not only product ID.

Snapshot ka matlab challan item mein product ka current name, SKU and unit price copy karke store karna. Kal product price change ho jaye, purana challan purane price par hi rehna chahiye.

```
Today:
Monitor price = 10000
Customer buys 5
Challan item stores unit_price_snapshot = 10000

Tomorrow:
Product master price changes to 12000
Old challan still shows 10000
```

8. Challan Tables

Table	Important columns
challans	id, challan_number, customer_id, total_quantity, total_amount, status, created_by, created_at
challan_items	id, challan_id, product_id, product_name, sku, quantity, unit_price, line_total

9. Status Transitions

DRAFT -> CONFIRMED

DRAFT -> CANCELLED

Usually blocked:

CONFIRMED -> DRAFT

CANCELLED -> CONFIRMED

10. Demo Script

- 1 Create a draft challan for a customer with two products and show stock does not change.
- 2 Confirm the challan and show stock reduced.
- 3 Open stock movement log and show OUT entries for each item.
- 4 Try confirming a challan with quantity greater than available stock and show proper error.
- 5 Change product price and show old challan still keeps old snapshot price.